6. Complex Power in AC Circuits

Introduction to Electric Circuits
Supplemental Instruction

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This document is not a substitute for a textbook or classroom instruction. It may contain errors and is intended to be used only to the extent that it is helpful to supplement students' learning experience.

1 Introduction

2 RMS Values

When we analyze the power of AC circuits, we use **RMS values**, which give the effective value of a sinusoidal voltage or current. For a sinusoidal voltage $\boldsymbol{V} = V_m\angle\alpha_v$ or current $\boldsymbol{I} = I_m\angle\alpha_i$, the RMS value is

$$V_{rms} = \frac{V_m}{\sqrt{2}} \qquad I_{rms} = \frac{I_m}{\sqrt{2}}$$

3 Average (True) Power

For a component or network, given $\boldsymbol{V} = V_m\angle\alpha_v$ and $\boldsymbol{I} = I_m\angle\alpha_i$, the **average power** is:

$$P = V_{rms}I_{rms}\cos(\alpha_v - \alpha_i)$$

Or, using magnitudes,

$$P = \frac{V_m I_m}{2}\cos(\alpha_v - \alpha_i)$$

Average power is sometimes also called **true power** or **real power**. The **unit** of average power is Watts (W). Average power is associated with resistance.

4 Reactive Power

For a component or network, given $\boldsymbol{V} = V_m\angle\alpha_v$ and $\boldsymbol{I} = I_m\angle\alpha_i$, the **reactive power** is:

$$Q = V_{rms}I_{rms}\sin(\alpha_v - \alpha_i)$$

Or, using magnitudes,

$$Q = \frac{V_m I_m}{2}\sin(\alpha_v - \alpha_i)$$

The **unit** of average power is Volt Amps Reactive (VAR). Reactive power is associated with reactive components (inductors and capacitors).

5 Apparent Power

For a component or network, given $\boldsymbol{V} = V_m\angle\alpha_v$ and $\boldsymbol{I} = I_m\angle\alpha_i$, the **apparent power** is:

$$S = V_{rms}I_{rms}$$

Or, using magnitudes,

$$S = \frac{V_m I_m}{2}$$

The **unit** of apparent power is Volt Amps (VA).

6 Power Factor

For a component or network, given $\boldsymbol{V} = V_m\angle\alpha_v$ and $\boldsymbol{I} = I_m\angle\alpha_i$, the **power factor** is:

$$PF = \cos(\alpha_v - \alpha_i)$$

Power factor is specified as either leading or lagging, describing whether the current leads or lags the voltage. If $\alpha_i > \alpha_v$, the power factor is leading; if $\alpha_i < \alpha_v$, the power factor is lagging.

7 Complex Power and the Power Triangle

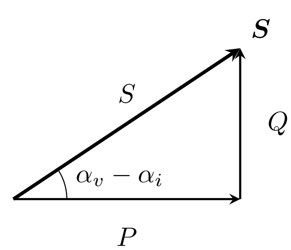
In summary, we have

$$P = V_{rms}I_{rms}\cos(\alpha_v - \alpha_i)$$
$$Q = V_{rms}I_{rms}\sin(\alpha_v - \alpha_i)$$
$$S = V_{rms}I_{rms}$$
$$PF = \cos(\alpha_v - \alpha_i)$$

Complex power is

$$\boldsymbol{S} = P + jQ = S\angle(\alpha_v - \alpha_i)$$

We can draw this as a triangle in the complex plane, where average power is the horizontal component, reactive power is the vertical component, and apparent power is the magnitude:



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